Can Early Training Dynamics Predict Out-of-Distribution Generalization?

AI2002 Artificial Intelligence (BDS-5A)

Project Phase 1: Problem Definition

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Track: A — Research & Development

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1. Problem Statement

Neural networks trained on data that contains a shortcut, meaning an easy feature that correlates with the label in the training set but not in deployment (for example, digit colour or image background), tend to rely on that shortcut instead of the real signal. Such a model can score highly on in-distribution data and still fail badly once the shortcut stops working. Recent work shows that this reliance appears early in training.

Existing methods for forecasting out-of-distribution (OOD) accuracy, such as Average Thresholded Confidence (ATC) and Agreement-on-the-Line (AGL), inspect only the finished model. Existing early-training work detects that a shortcut is being learned, but, to our knowledge, it has not been benchmarked as a quantitative predictor of final OOD accuracy against these static baselines.

This project investigates this gap.

Core question: If a model is observed for only the first 10–20% of training, can signals computed from that window predict its final accuracy on shifted data, and do they predict it better than ATC and AGL computed on the fully trained model?

Hypotheses

- **H1:** At least some early signals meaningfully predict final OOD accuracy.
- **H2:** Different early signals differ significantly in how predictive they are.
- **H3:** A combination of early signals outperforms the static, final-model baselines.
- **H4:** Reliable prediction is possible well before training finishes, potentially within the first 10–20% of the training budget.

2. Early-Training Signals and Baselines

Each trained model is treated as one data point: signals are measured on the model, and the target is its final OOD accuracy.

Signals are recorded at checkpoints at approximately 5%, 10%, 15%, and 20% of training.

Prediction Depth (PD)

Prediction Depth measures the number of layers after which a k-NN probe on hidden embeddings already matches the final prediction. Shortcut-driven examples are expected to be settled in early layers.

It will be computed per example and then summarised per model using measures such as mean prediction depth and early-layer share.

Used as: Early signal

Gradient-Starvation-Inspired Proxy

Gradient Starvation describes the tendency of a fast-learned feature to suppress the gradient available for other predictive features. Since the original work provides theory and a regulariser rather than a single ready-to-use metric for our forecasting task, we will define and document a measurable gradient-starvation-inspired proxy.

Possible measurements include the share of training loss and gradient behaviour attributable to shortcut-aligned samples.

Used as: Early signal

Feature-Ablation Reliance Score

This is our own score measuring the model's reliance on the shortcut feature.

It will be calculated as the drop in training-time accuracy when the shortcut feature is removed or randomised:

- colour neutralised on Colored MNIST;
- background replaced on Waterbirds.

Used as: Early signal

ATC — Average Thresholded Confidence

ATC learns a confidence threshold using source-domain data and predicts target accuracy as the fraction of unlabeled target examples whose confidence exceeds the learned threshold.

Used as: Static baseline

Agreement-on-the-Line (AGL)

Agreement-on-the-Line uses the relationship between in-distribution and OOD agreement across several trained models to estimate OOD accuracy without using target labels.

Used as: Static baseline

Fairness of Comparison

The static baselines may use unlabeled OOD inputs, while the early signals primarily use information available during training. We will explicitly state the information available to each method in the final report and, where possible, also report a variant of the early signals that incorporates unlabeled OOD data.

3. Paper Links

Core Papers

Baldock et al., "Deep Learning Through the Lens of Example Difficulty," NeurIPS 2021
Role: Prediction Depth
Link: https://arxiv.org/abs/2106.09647

Pezeshki et al., "Gradient Starvation: A Learning Proclivity in Neural Networks," NeurIPS 2021
Role: Gradient Starvation
Code: https://github.com/mpezeshki/Gradient_Starvation
Paper: https://arxiv.org/abs/2011.09468

Garg et al., "Leveraging Unlabeled Data to Predict Out-of-Distribution Performance," ICLR 2022
Role: ATC baseline
Code: https://github.com/saurabhgarg1996/ATC_code
Paper: https://arxiv.org/abs/2201.04234

Baek et al., "Agreement-on-the-Line: Predicting the Performance of Neural Networks under Distribution Shift," NeurIPS 2022
Role: AGL baseline
Code: https://github.com/kebaek/agreement-on-the-line
Paper: https://arxiv.org/abs/2206.13089

Arjovsky et al., "Invariant Risk Minimization," 2019
Role: Colored MNIST construction
Link: https://arxiv.org/abs/1907.02893

Sagawa et al., "Distributionally Robust Neural Networks for Group Shifts," ICLR 2020
Role: Waterbirds benchmark
Link: https://arxiv.org/abs/1911.08731

Closely Related Work

Murali et al.
Shows that shortcut learning can be observed through early training dynamics and Prediction Depth.

Yang et al., "Identifying Spurious Biases Early in Training through the Lens of Simplicity Bias," AISTATS 2024
Shows that spurious features can be learned within the early stages of training.
Link: https://proceedings.mlr.press/v238/yang24c

These works detect or analyse shortcut learning during training; our project instead investigates whether early training signals can quantitatively forecast final OOD accuracy and compares them against established static OOD prediction baselines.

4. Datasets

Colored MNIST — Controlled Setting

Digits are coloured so that colour correlates with the label with a tunable probability pp, following the construction of Arjovsky et al. (2019).

The dataset listed in our approved proposal is:

https://www.kaggle.com/datasets/youssifhisham/colored-mnist-dataset

Because shortcut strength must be varied systematically, the main experiments will use a version generated by ourselves from standard MNIST, with pp treated as an experimental parameter (for example, values from 0.5 to 0.99). The Kaggle copy will serve as a reference and sanity check.

The small size of Colored MNIST makes it feasible to conduct many training runs under different conditions.

Waterbirds — Realistic Validation

Waterbirds consists of bird images from CUB placed onto backgrounds from the Places dataset. The classification task is landbird versus waterbird, while the background acts as the spurious feature.

In the training set, approximately 95% of birds appear on their typical background, while the validation and test sets are balanced across bird and background combinations.

The dataset is available through WILDS and the original Group-DRO repository:

https://wilds.stanford.edu/

https://github.com/kohpangwei/group_DRO

https://github.com/p-lambda/wilds

Waterbirds has a fixed shortcut strength, so variation across models will come from different random seeds, architectures, and training hyperparameters.

5. Scope of the Project

Distribution Shifts Tested

Colored MNIST:

- Training colour-label correlation pp will be varied.
- OOD test sets will include:

    - reversed colour-label correlation;
    - colour uncorrelated with the label.

Waterbirds:

- A subpopulation shift from the biased training distribution to the group-balanced test distribution will be evaluated.
- Both average accuracy and worst-group accuracy will be reported.

Experimental Design

- A population of models will be trained for each dataset.
- For Colored MNIST, several values of pp, multiple random seeds, and small MLP and CNN architectures will be used.
- For Waterbirds, ResNet-based models will be evaluated across different seeds and training hyperparameters.
- Exact model counts will be fixed based on available computational resources.

For every model:

1. Early-training signals will be computed at the selected checkpoints.
2. Static baselines will be computed after training is complete.
3. The true final OOD accuracy will be recorded as the prediction target.

Regression models will then be trained to predict final OOD accuracy from the early signals. Both individual-signal models and a combined regularised linear regression model will be evaluated.

Evaluation Protocol

The methods will be evaluated using:

- Pearson correlation;
- Spearman correlation;
- Mean Absolute Error (MAE);
- Root Mean Squared Error (RMSE);
- R-squared (R2R^2).

Cross-validation will be used to evaluate prediction performance, including experiments with held-out shortcut strengths to test whether the forecasting method generalises across different levels of spurious correlation.

Prediction quality will also be evaluated as a function of the training checkpoint (5%, 10%, 15%, and 20%) to determine how early reliable forecasting becomes possible.

Bootstrap confidence intervals and paired comparisons against ATC and AGL will be used to assess differences between methods. Error analysis will be conducted to identify situations in which individual methods succeed or fail.

6. Out of Scope

The following are outside the scope of this project:

- Designing new debiasing or robust-training methods such as improved IRM or GroupDRO.
- Competing directly with robust-training methods.
- Language, tabular, or medical datasets.
- Very large models or pretraining large networks.
- Building a polished user interface. A UI may be developed only if time permits as a bonus component.

7. Deliverables

The project will produce:

- A reproducible codebase.
- A LaTeX technical report containing the methodology, replication-ready experimental protocol, results, error analysis, and challenges.
- Comparative experimental results showing which early-training signal, and how early in training, most reliably forecasts eventual OOD performance.
- Validation of the main findings across shortcut strengths on Colored MNIST and on the Waterbirds dataset.

8. Known Risks

- Agreement-on-the-Line may not perform reliably under strong spurious shifts. If this occurs, it will be reported as an experimental finding rather than repeatedly tuning the baseline until it performs well.
- Gradient Starvation does not provide a single ready-made metric for this forecasting setup. Therefore, the operational definition of the gradient-starvation-inspired proxy will be clearly documented, and alternative variants will be reported if necessary.
- The number of Waterbirds training runs may be limited by available computational resources; this limitation will be stated in the final report.